MAC F244, Semester I, AY 2026-2027

Project Component: Expectations and Evaluation Guidelines

1. Evaluation Scheme

The project component carries **20 marks** of the total course evaluation. The objective of the project is to explore an application of **stochastic calculus, stochastic processes, or financial mathematics** through mathematical modeling, numerical simulation, and computational implementation.

2. Project Requirements

Students are required to work in **groups of five members**. Each group may select a topic from the suggested list below or propose an alternative topic relevant to the course.

Project Proposal

Each group must submit **the idea of the project and the progress made on the same by 31st October 2026**, which must include:

- Project title and a brief description of the problem.
- Mathematical model or stochastic process to be investigated.
- Proposed computational methodology.
- Expected outcomes.
- Names and ID numbers of all group members.

Final Submission by 15th November

The final submission should include:

1. **Source Code:** Well-structured and properly commented code using R, Python, or another suitable programming language.
2. **Project Report:** A concise report of 3–4 pages describing the mathematical model, numerical methodology, results, and conclusions.
3. **Presentation/Viva:** A 15- to 20-minute viva on mathematical formulation and key findings.

3. Project Evaluation (20 Marks)

Evaluation Criterion	Marks
Report	5
Mathematical formulation and explanation of model	5
Numerical results, analysis, and interpretation	5
Viva/Presentation	5
Total	**20**

4. Suggested Project Ideas

1. Develop a trading strategy for stock/or derivative using Geometric Brownian Motion.
2. Develop a trading strategy for stock/or derivative using mean-reverting stochastic processes: Ornstein–Uhlenbeck and CIR Models.
3. Estimation of volatility and covariance using historical financial market data for different SDE price models; use the same for portfolio optimization and compare the results.
4. Modeling correlated financial assets using multidimensional Brownian motion. Use the same to develop a trading strategy.
5. Brownian Motion or GBM and First-Passage Time Problems in Financial Mathematics.
6. Application of numerical solutions of stochastic differential equations using Euler–Maruyama and Milstein Methods.
7. European Option Pricing: Comparison of Binomial Trees and the Black–Scholes Model.
8. Monte Carlo Pricing of European Options Under Risk-Neutral Dynamics.
9. Pricing Asian and Lookback Options Using Monte Carlo Simulation.
10. Pricing Barrier Options Using Brownian Motion.
11. American Option Pricing Using Binomial Trees and Backward Induction.
12. Delta Hedging of European Options and Analysis of Hedging Errors.
13. Portfolio Risk and Diversification Under Correlated Asset Dynamics.
14. Continuous-Time Portfolio Optimization: The Merton Problem.
15. Monte Carlo Estimation of Value at Risk and Expected Shortfall.
16. Optimal Stopping Problems and Their Applications to American Option Pricing.
17. Pairs Trading Strategies Based on Mean-Reverting Stochastic Processes.

5. General Instructions

- Students may use R, Python, or any other suitable programming language.
- Projects should demonstrate a clear connection between mathematical theory and computational implementation.
- Students are encouraged to use real financial market data wherever appropriate.
- Numerical experiments should investigate the effects of changes in data or assets.
- Simply reproducing existing formulas or numerical results without meaningful analysis will not be sufficient.
- All mathematical models, algorithms, and datasets must be appropriately referenced.
- Originality, mathematical understanding, quality of implementation, and interpretation of results will be considered during evaluation.